



Department of Computer Science

Assignment 04

Course Code: CS-112

Name: _____

Course Title: Object Oriented Programming

Registration No. _____

Total Marks: 20

Discipline/Semester/Section: BSCS 2nd A/B

Assigned Date: 04-06-2025

Deadline:12-06-2025

Submission Instructions:

- ✓ Plagiarism will result in zero marks.
 - ✓ Add comments in each part of the code to explain functionality.
 - Submit the assignment in the following format:**
 - ✓ Take a screenshot of your Java code, Package Structure and output.
 - ✓ Create separate classes (open new java class).
 - ✓ Use meaningful class, methods, and variable names and Packages Names etc.
 - ✓ Compile everything into a Word file/pdf and submit it on google classroom with code.
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(CLO-3, GA-4, C4)

Q1.

The University of Wah is developing a CGPA Calculator system to help students calculate their academic performance.

The system should allow students to input their name, registration number, number of semesters completed and course details for each semester that includes number of subjects in a particular semester (e.g. Semester 1: Subjects 5), grade points and credit hours for each subject

Decompose the system requirements to design a solution that calculates the GPA for each semester and uses those to compute the CGPA across all semesters.

Results should be displayed in a dialog showing the student's name, registration number, semester-wise GPA, and the final CGPA.

All invalid inputs should be handled using proper exception handling.

To maintain scalability and clarity, the application is divided into the following packages:

Packages:

- model – contains student-related data models (e.g., Student, Semester, Subject)
- service – handles GPA/CGPA calculation logic
- UI – manages the GUI using Swing components
- main – contains the application launcher

Assignment Rubrics

Criteria	Exemplary	Good	Satisfactory	Poor / Incomplete	Max Marks
1. Correctness & Functionality	Fully functional; all GPA/CGPA calculations are accurate; input validation works.	Mostly functional; minor logic bugs.	Basic functionality works; issues in edge cases.	Major parts broken or missing.	6
2. GUI Design & Usability	Clean layout; easy to use; good use of Swing components.	Usable; small issues in layout or flow.	Usable but unintuitive or cluttered.	Very poor or no GUI.	4
3. Code Structure & Packaging	Excellent use of packages and class organization.	Mostly correct use of packages.	Some structure; not modular.	No package structure; all code in one file.	4
4. Exception Handling	All errors well-handled with proper messages.	Minor exceptions not handled.	Basic validation done.	No error handling; crashes on bad input.	3
5. Code Quality & Readability	Clean, consistent, and readable code; uses OOP well.	Readable with small inconsistencies.	Understandable, but messy or repetitive.	Poorly written; lacks comments or structure.	3
Total					20 Marks